INFOSYS SE PSEUDOCODE VAULT • LOOPS & CONDITIONAL TRACING

While, For, Nested Loops, Break/Continue, Step Modifications • Total Solved Questions: 74

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: For Loop with Condition

Q1. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 10 step 1 do
if (i % 3 != 0)
s = s + i
end-if
end-for
display s
```

(A) 30
(B) 37
(C) 42
(D) 45

Correct Answer: Option (B) : 37

■ **Step-by-Step Logic:** Total 1..10=55; multiples of 3 are 3+6+9=18; 55–18=37.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop

Q2. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 5 step 1 do
if (i + j == 6)
c = c + 1
end-if
end-for
end-for
display c
```

(A) 2
(B) 3
(C) 4
(D) 5

Correct Answer: Option (C) : 4

■ **Step-by-Step Logic:** Pairs are (1,5),(2,4),(3,3),(4,2): 4.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop Sum

Q3. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 3 step 1 do
for j = 1 to i + 1 step 1 do
s = s + j
end-for
end-for
display s
```

(A) 17
(B) 18
(C) 19
(D) 20

Correct Answer: Option (C) : 19

■ **Step-by-Step Logic:** For i=1,2,3 the inner sums are 3,6,10; total=19. Infosys Medium Pseudocode Practice • Page 36

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: While Loop

Q4. What will be the output of the following pseudocode?

```
Set Integer x = 5
Set Integer s = 0
while (x <= 25)
s = s + x
x = x + 5
end-while
display s
```

(A) 50
(B) 60
(C) 75
(D) 80

Correct Answer: Option (C) : 75

■ **Step-by-Step Logic:** 5+10+15+20+25=75.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Break in Nested Loop

Q5. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 4 step 1 do
if (j == 3)
break
end-if
c = c + 1
end-for
end-for
display c
```

(A) 4
(B) 6
(C) 8
(D) 12

Correct Answer: Option (C) : 8

■ **Step-by-Step Logic:** The inner loop adds 1 for j=1 and j=2, then breaks at j=3. 4×2=8.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Continue

Q6. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 8 step 1 do
if (i % 3 == 0)
continue
end-if
s = s + i
end-for
display s
```

(A) 24
(B) 27
(C) 30
(D) 33

Correct Answer: Option (B) : 27

■ **Step-by-Step Logic:** 1..8 sum=36; skip 3 and 6, so 36–9=27.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Reverse Loop

Q7. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 12 to 2 step -2 do
s = s + i / 2
end-for
display s
```

(A) 18
(B) 21
(C) 24
(D) 27

Correct Answer: Option (B) : 21

■ **Step-by-Step Logic:** Values of i/2: 6,5,4,3,2,1; sum=21.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop with Changing Step

Q8. What will be the output of the following pseudocode?

```
Set Integer i = 1
Set Integer s = 0
while (i < 20)
s = s + i
i = i * 2 + 1
end-while
display s
```

(A) 24
(B) 26
(C) 28
(D) 30

Correct Answer: Option (B) : 26

■ **Step-by-Step Logic:** i values are 1,3,7,15; sum=26.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop Count

Q9. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 5 step 1 do
for j = i to 5 step 1 do
c = c + 1
end-for
end-for
display c
```

(A) 10
(B) 12
(C) 15
(D) 20

Correct Answer: Option (C) : 15

■ **Step-by-Step Logic:** Counts are 5+4+3+2+1=15.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop with Modulo

Q10. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 2 to 14 step 2 do
if (i % 4 == 0)
s = s + i
end-if
end-for
display s
```

(A) 12
(B) 16
(C) 20
(D) 24

Correct Answer: Option (D) : 24

■ **Step-by-Step Logic:** Multiples of 4 are 4,8,12; sum=24.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Conditional Loop

Q11. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 5 step 1 do
if (i % 2 == 1)
for j = 1 to i step 2 do
s = s + j
end-for
end-if
end-for
display s
```

(A) 12
(B) 14
(C) 16
(D) 18

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** For i=1,3,5 the inner sums are 1,4,9; total=14. Infosys Medium Pseudocode Practice • Page 37

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop State

Q12. What will be the output of the following pseudocode?

```
Set Integer x = 3
for i = 1 to 4 step 1 do
x = x + i
x = x - 1
end-for
display x
```

(A) 8
(B) 9
(C) 10
(D) 11

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** Each iteration changes x by i–1. Total change=0+1+2+3=6; final x=9.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop Product

Q13. What will be the output of the following pseudocode?

```
Set Integer p = 1
for i = 2 to 8 step 2 do
p = p * (i / 2)
end-for
display p
```

(A) 12
(B) 24
(C) 36
(D) 48

Correct Answer: Option (B) : 24

■ **Step-by-Step Logic:** Factors are 1,2,3,4; product=24.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: While + Modulo

Q14. What will be the output of the following pseudocode?

```
Set Integer n = 29
Set Integer c = 0
while (n > 0)
if (n % 2 == 1)
c = c + 1
end-if
n = n / 2
end-while
display c
```

(A) 2
(B) 3
(C) 4
(D) 5

Correct Answer: Option (B) : 3

■ **Step-by-Step Logic:** Binary 29 is 11101, containing four 1s.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop Arithmetic

Q15. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 3 step 1 do
for j = 1 to 3 step 1 do
s = s + i * j
end-for
end-for
display s
```

(A) 12
(B) 18
(C) 24
(D) 36

Correct Answer: Option (D) : 36

■ **Step-by-Step Logic:** The double sum factorizes: (1+2+3)×(1+2+3)=6×6=36.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Break with Sum

Q16. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 2 to 12 step 2 do
s = s + i
if (s > 15)
break
end-if
end-for
display s
```

(A) 12
(B) 18
(C) 20
(D) 24

Correct Answer: Option (B) : 18

■ **Step-by-Step Logic:** Running sums: 2,6,12,20; break after 20.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Continue with Product

Q17. What will be the output of the following pseudocode?

```
Set Integer p = 1
for i = 1 to 6 step 1 do
if (i % 2 == 0)
continue
end-if
p = p * i
end-for
display p
```

(A) 9
(B) 12
(C) 15
(D) 20

Correct Answer: Option (C) : 15

■ **Step-by-Step Logic:** Odd values 1,3,5 are multiplied: 1×3×5=15.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop with Shift

Q18. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 4 step 1 do
s = s + (i << 1)
end-for
display s
```

(A) 16
(B) 18
(C) 20
(D) 24

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** 2+4+6+8=20.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop with Right Shift

Q19. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 8 to 32 step 8 do
s = s + (i >> 2)
end-for
display s
```

(A) 15
(B) 18
(C) 20
(D) 24

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** The shifted values are 2,4,6,8; sum=20. Infosys Medium Pseudocode Practice • Page 38

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop with Break

Q20. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 4 step 1 do
c = c + 1
if (c == 7)
break
end-if
end-for
end-for
display c
```

(A) 7
(B) 8
(C) 12
(D) 13

Correct Answer: Option (D) : 13

■ **Step-by-Step Logic:** The counter continues across outer iterations. It reaches 7, breaks the current inner loop, then continues; 4 more inner iterations occur, so final c=13.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop and Comparison

Q21. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 9 step 1 do
if (i * i < 50)
c = c + 1
end-if
end-for
display c
```

(A) 6
(B) 7
(C) 8
(D) 9

Correct Answer: Option (B) : 7

■ **Step-by-Step Logic:** i=1..7 satisfy $i^2<50$; count=7.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function with Loop

Q22. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer s = 1
for i = 1 to n step 1 do
s = s + i
end-for
return s
End function
display f(6)
```

(A) 20
(B) 21
(C) 22
(D) 23

Correct Answer: Option (C) : 22

■ **Step-by-Step Logic:** 1+(1+2+3+4+5+6)=22.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Condition

Q23. What will be the output of the following pseudocode?

```
Integer f(Integer n)
if (n % 4 == 0)
return n / 2
else
return n + 4
end-if
End function
display f(14)
```

(A) 9
(B) 14
(C) 18
(D) 28

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** 14 is not divisible by 4, so 14+4=18. Infosys Medium Pseudocode Practice • Page 42

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Digit Logic

Q24. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer s = 0
while (n > 0)
s = s + n % 10
n = n / 10
end-while
return s
End function
display f(386)
```

(A) 14
(B) 15
(C) 16
(D) 17

Correct Answer: Option (D) : 17

■ **Step-by-Step Logic:** 3+8+6=17.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Nested

Q25. What will be the output of the following pseudocode?

```
Integer g(Integer x)
return x * 2
End function
Integer f(Integer x)
return g(x) + g(x+1)
End function
display f(4)
```

(A) 16
(B) 18
(C) 20
(D) 22

Correct Answer: Option (B) : 18

■ **Step-by-Step Logic:** g(4)=8 and g(5)=10; total=18.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Loop Product

Q26. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer p = 1
for i = 1 to n step 1 do
if (i % 2 == 0)
p = p * i
end-if
end-for
return p
End function
display f(6)
```

(A) 24
(B) 36
(C) 48
(D) 60

Correct Answer: Option (C) : 48

■ **Step-by-Step Logic:** Multiply even values 2×4×6=48.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function with Loop and Shift

Q27. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer s = 0
for i = 1 to n step 1 do
s = s + (i << 1)
end-for
return s
End function
display f(5)
```

(A) 20
(B) 25
(C) 30
(D) 35

Correct Answer: Option (C) : 30

■ **Step-by-Step Logic:** 2×(1+2+3+4+5)=30.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Loop + XOR

Q28. What will be the output of the following pseudocode?

```
Set Integer x = 0
for i = 1 to 5 step 1 do
x = x ^ i
end-for
display x
```

(A) 0
(B) 1
(C) 2
(D) 3

Correct Answer: Option (A) : 0

■ **Step-by-Step Logic:** 0^1=1, ^2=3, ^3=0, ^4=4, ^5=1.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Loop + AND

Q29. What will be the output of the following pseudocode?

```
Set Integer x = 15
for i = 1 to 3 step 1 do
x = x & (14 - i)
end-for
display x
```

(A) 8
(B) 10
(C) 12
(D) 14

Correct Answer: Option (B) : 10

■ **Step-by-Step Logic:** x=15&13=13; 13&12=12; 12&11=8.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Nested Loop + Modulo

Q30. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 4 step 1 do
if ((i + j) % 3 == 0)
c = c + 1
end-if
end-for
end-for
display c
```

(A) 4
(B) 5
(C) 6
(D) 7

Correct Answer: Option (B) : 5

■ **Step-by-Step Logic:** Pairs with sum divisible by 3: 5 pairs. Correct answer B.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Nested Conditions

Q31. What will be the output of the following pseudocode?

```
Set Integer a = 9
Set Integer b = 6
Set Integer c = 3
if (a > b)
if ((a - b) == c)
display a + b + c
else
display a - b
end-if
else
display b + c
end-if
```

(A) 12
(B) 15
(C) 18
(D) 21

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** 9>6 and 9−6=3, so the nested condition is true; 9+6+3=18.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q32. How many times does the loop body run, and

```
what is printed?
Integer
i, count
Set
i = 1, count = 0
while
i < 20 do
i
= i * 2
count
= count + 1
end
while
Print
count
```

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: Option (C) : 5

■ **Step-by-Step Logic:** Trace i and count: start i=1. Pass 1: i=2, count=1. Pass 2: i=4, count=2. Pass 3: i=8, count=3. Pass 4: i=16, count=4. Pass 5: i=32, count=5. Now i=32 is not less than 20, so the loop stops. Printed value is 5.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q33. What will be the output of the following pseudocode?

```
What is the output of the following Pseudo
Code?
Integer
x, i
Set
x = 0
for
i = 1 to 6 do
if
i % 2 == 0 then
continue
end
if
x
= x + i
end
for
Print
x
```

(A) 9
(B) 12
(C) 16
(D) 21

Correct Answer: Option (A) : 9

■ **Step-by-Step Logic:** continue skips even values of i. The loop runs i = 1 to 6, so only the odd values 1, 3 and 5 are added. x = 1 + 3 + 5 = 9.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q34. What will be the output of the following pseudocode?

```
What is printed by the following Pseudo
Code?
Integer
i, j, count
Set
count = 0
for
i = 1 to 4 do
for
j = 1 to i do
count
= count + 1
end
for
end
for
Print
count
```

(A) 4
(B) 8
(C) 10
(D) 16

Correct Answer: Option (C) : 10

■ **Step-by-Step Logic:** The inner loop runs i times for each i from 1 to 4. Total iterations = 1+2+3+4 = 10. So count prints 10.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q35. What will be the output of the following pseudocode?

```
A queue follows FIFO. What is printed by the
following Pseudo Code?
Queue
Q
enqueue(Q,
2)
enqueue(Q,
4)
enqueue(Q,
6)
dequeue(Q)
enqueue(Q,
8)
Print
front(Q)
```

(A) 2
(B) 4
(C) 6
(D) 8

Correct Answer: Option (B) : 4

■ **Step-by-Step Logic:** After three enqueues the queue is [2, 4, 6] with 2 at the front. dequeue removes 2. enqueue 8 makes [4, 6, 8]. front returns 4.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q36. What will be the output of the following pseudocode?

```
A stack follows LIFO. What is printed by the
following Pseudo Code?
Stack
S
push(S,
5)
push(S,
8)
push(S,
3)
pop(S)
push(S,
7)
Print
peek(S)
```

(A) 3
(B) 5
(C) 7
(D) 8

Correct Answer: Option (C) : 7

■ **Step-by-Step Logic:** Stack after the three pushes: [5, 8, 3] with 3 on top. pop removes 3. push 7 makes the stack [5, 8, 7]. peek returns the top element, which is 7.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q37. What will be the output of the following pseudocode?

```
What is printed?
Integer
n, d
Set
n = 48, d = 0
while
n > 1 do
n
= n / 2
d
= d + 1
end
while
Print
d
```

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: Option (C) : 5

■ **Step-by-Step Logic:** 48→24→12→6→3→1. That is five divisions. d = 5. (Integer division: 3/2=1.)

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q38. What will be the output of the following pseudocode?

```
What is printed?
Integer
A[5] = {2, 4, 6, 8, 10}
Integer
i, s
Set
s = 0
for
i = 0 to 4 do
if
A[i] % 4 == 0 then s = s + A[i]
end
for
Print
s
```

(A) 12
(B) 18
(C) 24
(D) 30

Correct Answer: Option (A) : 12

■ **Step-by-Step Logic:** Values divisible by 4 are 4 and 8. s = 4 + 8 = 12.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q39. What will be the output of the following pseudocode?

```
What does G(4) return?
Function
G(n)
if
n <= 1 then return 1
return
n * G(n - 1)
End
Function
Print
G(4)
```

(A) 10
(B) 16
(C) 24
(D) 64

Correct Answer: Option (C) : 24

■ **Step-by-Step Logic:** G is factorial. G(4)=4×3×2×1=24.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q40. What will be the output of the following pseudocode?

```
[Loops]
What is printed?
Integer
i, s
Set
s = 0
for
i = 1 to 10 do
if
i % 2 == 1 then s = s + i
end
for
Print
s
```

(A) 25
(B) 30
(C) 45
(D) 55

Correct Answer: Option (A) : 25

■ **Step-by-Step Logic:** The loop adds the odd values 1+3+5+7+9 = 25.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q41. What will be the output of the following pseudocode?

```
[Loops]
What is printed?
Integer
i, p
Set
p = 1
for
i = 1 to 4 do
p
= p * i
end
for
Print
p
```

(A) 10
(B) 16
(C) 24
(D) 120

Correct Answer: Option (C) : 24

■ **Step-by-Step Logic:** p multiplies 1×2×3×4 = 24.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q42. What will be the output of the following pseudocode?

```
What
does R(6) return?
Function
R(n)
if
n < 3 then return n
return
R(n - 1) + 2
End
Function
Print
R(6)
```

(A) 6
(B) 8
(C) 10
(D) 12

Correct Answer: Option (C) : 10

■ **Step-by-Step Logic:** R(2)=2. Each later call adds 2. R(6)=2 + 2×4 = 10.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q43. What will be the output of the following pseudocode?

```
What
is printed?
Integer
i, c
Set
c = 0
for
i = 1 to 8 do
if
i % 2 == 0 then c = c + i
end
for
Print
c
```

(A) 16
(B) 18
(C) 20
(D) 36

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** Even values 2+4+6+8 = 20.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q44. What will be the output of the following pseudocode?

```
What
does U(4) return?
Function
U(n)
if
n == 0 then return 1
return
n + U(n - 1)
End
Function
Print
U(4)
```

(A) 8
(B) 9
(C) 11
(D) 24

Correct Answer: Option (C) : 11

■ **Step-by-Step Logic:** U(0)=1, then +1+2+3+4 = 11.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q45. What will be the output of the following pseudocode?

```
What
is printed?
Integer
i, n
Set
n = 0
for
i = 1 to 6 do
if
i % 3 == 0 then n = n + i
end
for
Print
n
```

(A) 6
(B) 9
(C) 12
(D) 21

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** Multiples of 3 from 1 to 6 are 3 and 6. Sum = 9.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q46. What will be the output of the following pseudocode?

```
What
is printed?
Integer
i, t
Set
t = 1
for
i = 1 to 5 do
t
= t * 2
end
for
Print
t
```

(A) 10
(B) 16
(C) 32
(D) 64

Correct Answer: Option (C) : 32

■ **Step-by-Step Logic:** Start 1, multiply by 2 five times: 32.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q47. What will be the output of the following pseudocode?

```
What
does W(5) return?
Function
W(n)
if
n <= 1 then return 2
return
W(n - 1) + 3
End
Function
Print
W(5)
```

(A) 11
(B) 14
(C) 15
(D) 17

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** W(1)=2. Add 3 four times: 2+12=14.

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q48. 1. What will be the output of the following pseudo code?

```
Input m = 9,n = 6,
m= m + 1;
n= n - 1;
m= m + n;
if(m>n)
print m
else
print n
```

(A) 5
(B) 7
(C) 9
(D) 12

Correct Answer: Option (A) : 5

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (A) : 5.

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q49. 1. What will be the output of the following pseudocode?

```
Integer i
Set i = 3
do
print i + 3
i = i - 1
while(i not equals 0)
end while
```

(A) 6 6 6
(B) 6 5 6
(C) 5 5 5
(D) 6 5 4

Correct Answer: Option (D) : 6 5 4

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (D) : 6 5 4.

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q50. What will be the output of the following pseudocode?

```
1.     If the sequence of operations are performed on a
stack, the sequence of pop will be?

Push(1), Push(2), Pop(), Push(1), Push(2), Pop(), Pop(),
Pop(), Push(2), Pop().
```

(A) 2 2 1 1 2
(B) 2 2 1 2 2
(C) 2 1 2 2 1
(D) 2 1 2 2 2

Correct Answer: Option (A) : 2 2 1 1 2

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (A) : 2 2 1 1 2.

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q51. What will be the output of the following pseudocode?

```
T The concatenation of two lists is to be performed in
O(1) time. Which of the following implementation of a
list should be used?
```

(A) a) Singly linked list
(B) a) Doubly linked list
(C) a) Circular doubly linked list
(D) a) Array implementation of list

Correct Answer: Option (C) : a) Circular doubly linked list

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (C) : a) Circular doubly linked list.

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q52. What will be the output of the following pseudocode?

```
1.     Find the output of the following pseudo code.
Integer x,y,z
Set x=9, y=2, z=1
if ( x+y > 8)
if ( 7> x+z)
z=7
End if
If ( 9> x-z)
z=9
End if
y=y+z+1
End if

Print x+y+z
```

(A) 31
(B) 11
(C) 42
(D) 30

Correct Answer: Option (D) : 30

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (D) : 30.

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q53. How many times campus monk will be printed? Set k = 1, p = 4 Int j = p + k / 2 If (j equals p) Print "campus monk" End if j = p + k / 5 If (j not equal p) Print "campus monk" End if Print "campus monk"

(A) 1 Times
(B) 2 Times
(C) 3 Times
(D) 4 Times

Correct Answer: Option (B) : 2 Times

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (B) : 2 Times.

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q54. What will be the output of the following pseudocode? Initial values: a = 4, b = 4, c = 7 Int fun(int a, int b, int c) { For (each c from 3 to 5) { a = (c + c) ^ b; If ((a + c) < (c + a)) { b = (a + 11) + c; } Else { c = b + b; a = 3 + b; continue; } } Return a + b; }

(A) 10
(B) 12
(C) Error
(D) 11

Correct Answer: Option (D) : 11

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (D) : 11.

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q55. What will be the output of the following pseudocode?

```
Predict the output of the given pseudocode if the value of the number is 6: Read number k = 2 i = 2 while i <= number: k = k * i i = i + 1 end while write k
```

(A) 1440.0
(B) 1340.0
(C) 1700.0
(D) 1560.0

Correct Answer: Option (A) : 1440.0

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (A) : 1440.0.

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q56. What will be the output of the following pseudocode?

```
Q1.) Integer p, q, r Set p = 0, q = 2, r = 9 r = 7 + p q = q + r for (each r from 4 to 7) p = (p + p) & q if ((p + q) < (r - p) || 8 < p) p = (p + 2) + q Jump out of the loop End if End for Print p + q
```

(A) 5
(B) 9
(C) 19
(D) 20

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (B) : 9.

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q57. What will be the output of the following pseudocode?

```
Q3.) Integer value, n Set value = 1, n = 45 while (value less than equal to n) value = value << 1 end loop Print value
```

(A) 16
(B) 32
(C) 64
(D) 36

Correct Answer: Option (C) : 64

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (C) : 64.

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q58. What will be the output of the following pseudocode?

```
Q6. What is the output for p=15 and q=4. Integer solve (Integer p, Integer q) Integer value while(q) value = p MOD q p = q q = value End while return p End function solve()
```

(A) 3
(B) 1
(C) 2
(D) 0

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (B) : 1.

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q59. What will be the output of the following pseudocode?

```
Q9. What is the output for x = 10, y = 5 Integer solve(Integer x, Integer y) if(y > 0) if(x > 0) return x + y + solve(2, y - 5) + solve(x - 10, 1) End if End if return x + y End function
```

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q60. What will be the output of the following pseudocode?

```
Q10. What is the output of following Integer value, n Set value = 32, n = 1 while(value >= n) value = value >> 1 end loop Print value
```

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q61. What will be the output of the following pseudocode?

```
What is the output of following pseudo code: Fun() { Int a=10 , b=20 , c=30; If(c>b>a) Print(“True”); Else Print(“False”); return 0; }
```

(A) True
(B) False
(C) Error
(D) -1

Correct Answer: Option (B) : False

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (B) : False.

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q62. What will be the output of the following pseudocode?

```
What is the output of following: Integer x, y Set x = 15, y = 12 y = x – 1 do{ Print x x = y + (x – 2) } while(x < 40) end do while
```

(A) 14 26
(B) 15 27 39
(C) 27 39
(D) 15

Correct Answer: Option (B) : 15 27 39

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (B) : 15 27 39.

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q63. What will be the output of the following pseudocode?

```
What is the output if fun(5) is called? Function fun(n)
If n == 0 Return 0 Else Print(n) Return fun(n + 1) End If
End Function
```

(A) 1 2 3 4 5
(B) 5 4 3 2 1
(C) Stack overflow
(D) -1

Correct Answer: Option (C) : Stack overflow

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (C) : Stack overflow.

[College ITM Mock Paper 1]

Topic: Infosys SE College Mock Test

Q64. What will be the output of the following pseudocode?

```
What is printed?
x = 45; count = 0;
while (x != 0) {
        x = x & (x - 1);
        count++;
}
print(count ^ (45 >> 2))
```

(A) 4
(B) 11
(C) 15
(D) 7

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 1]

Topic: Infosys SE College Mock Test

Q65. What will be the output of the following pseudocode?

```
What is printed?
s = 0
for i = 1 to 5
        for j = i to 5
                if ((i + j) % 3 == 0) continue
                if (j == 5) break
                s = s + i * j
        end for
end for
print(s)
```

(A) 38
(B) 42
(C) 46
(D) 50

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 2]

Topic: Infosys SE College Mock Test

Q66. What will be the output of the following pseudocode?

```
What is printed? (x and y are unsigned integers)
x = 58; y = 0;
while (x > 0) {
        y = (y << 1) | (x & 1);
        x = x >> 1;
}
print(y)
```

(A) 23
(B) 29
(C) 37
(D) 41

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 2]

Topic: Infosys SE College Mock Test

Q67. What will be the output of the following pseudocode?

```
What is printed?
s = 0
for i = 1 to 4
        for j = 1 to 4
                if (i == j) continue
                if (i + j > 6) break
                s = s + (i * j)
        end for
end for
print(s)
```

(A) 40
(B) 44
(C) 46
(D) 48

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 2]

Topic: Infosys SE College Mock Test

Q68. What will be the output of the following pseudocode?

```
A queue is used (FIFO). What is printed?
enqueue(2); enqueue(4); enqueue(6);
a = dequeue();
enqueue(a + 10);
b = dequeue();
c = dequeue();
print(b + c)
```

(A) 10
(B) 16
(C) 18
(D) 22

Correct Answer: Option (A) : 10

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (A) : 10.

[College ITM Mock Paper 3]

Topic: Infosys SE College Mock Test

Q69. What will be the output of the following pseudocode?

```
A stack is used (LIFO). What is printed?
```

(A) 39
(B) 33
(C) 42
(D) 30

Correct Answer: Option (A) : 39

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches Option (A) : 39.

[College ITM Mock Paper 4]

Topic: Infosys SE College Mock Test

Q70. What will be the output of the following pseudocode?

```
This algorithm reverses the first k elements of a queue
using a stack. What is printed?
queue = [1, 2, 3, 4, 5]
stack = []
k = 3
for i = 1 to k
        push(stack, dequeue(queue))
end for
while (stack is not empty)
        enqueue(queue, pop(stack))
end while
for i = 1 to (5 - k)
        enqueue(queue, dequeue(queue))
end for
print(queue)
```

(A) [3, 2, 1, 4, 5]
(B) [1, 2, 3, 4, 5]
(C) [5, 4, 3, 2, 1]
(D) [3, 2, 1, 5, 4]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 5]

Topic: Infosys SE College Mock Test

Q71. What will be the output of the following pseudocode?

```
What is printed? (>>=right shift, <<=left shift, &=AND,
^=XOR)
x = 0;
for i = 0 to 7
        if (((45 >> i) & 1) == 1)
                x = x + (1 << (7 - i))
        end if
end for
y = x ^ 45
print(y)
```

(A) 153
(B) 180
(C) 135
(D) 171

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 6]

Topic: Infosys SE College Mock Test

Q72. What will be the output of the following pseudocode?

```
What is printed?
s = 0
for i = 1 to 5
        for j = 1 to 5
                if ((i + j) % 4 == 0) continue
                if (j > i + 1) break
                s = s + i * j
        end for
end for
print(s)
```

(A) 142
(B) 138
(C) 146
(D) 150

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 6]

Topic: Infosys SE College Mock Test

Q73. The following sorts a stack using a temporary stack (the top of a stack is its last element). How many

```
times is an element moved from tmp back to st?
st = [3, 1, 4, 2]; tmp = []; moves = 0
while (st is not empty)
        t = pop(st)
        while (tmp is not empty and top(tmp) > t)
                push(st, pop(tmp))
                moves = moves + 1
        end while
        push(tmp, t)
end while
print(moves)
```

(A) 3
(B) 2
(C) 4
(D) 5

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.

[College ITM Mock Paper 6]

Topic: Infosys SE College Mock Test

Q74. What will be the output of the following pseudocode?

```
What is printed? (dynamic programming for the minimum
number of coins; INF is a very large number)
coins = [1, 3, 4]
dp[0] = 0; dp[1..6] = INF
for i = 1 to 6
        for each c in coins
                if (i >= c)
                        dp[i] = min(dp[i], dp[i - c] + 1)
                end if
        end for
end for
print(dp[6])
```

(A) 2
(B) 3
(C) 4
(D) 6

Correct Answer: See Logic

■ **Step-by-Step Logic:** Evaluating the pseudocode step-by-step confirms the evaluated result matches See Logic.